0580_m24_ms_42_Q8

Question	Answers	Marks	Partial Marks
8(a)	Ruled line with negative gradient and positive y-intercept	**2**	**B1** for ruled line with negative gradient or for ruled line with positive y-intercept or straight line with negative gradient and positive y-intercept
8(b)	Negative quadratic, with vertex at origin	**2**	**B1** for negative quadratic in other position or for sketch in 3rd and 4th quadrants only with single maximum at (0, 0) and no other turning point or for positive quadratic, with vertex at origin
8(c)(i)	$18x - 6x^2$ isw	**B2**	**B1** for one correct term $18x$ or $-6x^2$ seen
	setting *their* derivative = 0 or $\frac{dy}{dx} = 0$	**M1**	**Dep** on at least B1 earned or *their* derivative = $\pm 18x \pm 6x^2$
	(0, 10) and (3, 37)	**B2**	**B1** for $x = 0$ and $x = 3$ or for (0, 10) or (3, 37)

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Question	Answer	Marks	Partial Marks
8(c)(ii)	(0, 10) minimum with correct reason AND (3, 37) maximum with correct reason	**3**	Reasons could be e.g. 1 A reasonable sketch of a negative cubic 2 Correct use of 2nd derivative = –12(0) + 18 = 18, 18 > 0, so (0, 10) is a minimum oe. 2nd derivative = –12(3) + 18 = –18, –18 < 0 so (3, 37) is a maximum oe. 3 Evaluates correctly values of *y* on both sides of both correct stationary points 4 Finds gradient on each side of both correct stationary points. **B2** for 1 correct with correct reason for that stationary point or for **both** *x*-values correct and reasonable sketch of a negative cubic, or for correct substitution and evaluation of **both** of *their* *x*-values into *their* second derivative or substitution and evaluation for one *x*-value on both sides of **both** of *their* stationary points to find the gradients soi or **M1** for showing [2nd derivative =] $-12x + 18$ or correct FT *their* 2nd derivative or substitution and evaluation shown for one *x*-value on both sides of **one** of *their* stationary points to find the gradients soi or for sketch of any negative cubic.